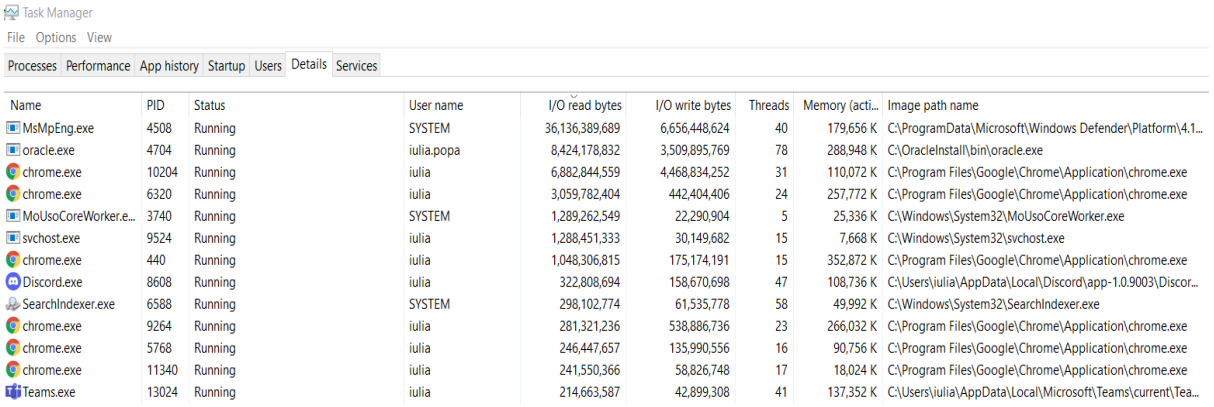
344C5
Popa Iulia

Laboratory 8 - I/O Monitoring (Windows)

Task 01. Task Manager



Name	PID	Status	User name	I/O read bytes	I/O write bytes	Threads	Memory (acti...	Image path name
MsMpEng.exe	4508	Running	SYSTEM	36,136,389,689	6,656,448,624	40	179,656 K	C:\ProgramData\Microsoft\Windows Defender\Platform\4.1...
oracle.exe	4704	Running	iulia.popa	8,424,178,832	3,509,895,769	78	288,948 K	C:\OracleInstall\bin\oracle.exe
chrome.exe	10204	Running	iulia	6,882,844,559	4,468,834,252	31	110,072 K	C:\Program Files\Google\Chrome\Application\chrome.exe
chrome.exe	6320	Running	iulia	3,059,782,404	442,404,406	24	257,772 K	C:\Program Files\Google\Chrome\Application\chrome.exe
MoUsoCoreWorker.e...	3740	Running	SYSTEM	1,289,262,549	22,290,904	5	25,336 K	C:\Windows\System32\MoUsoCoreWorker.exe
svchost.exe	9524	Running	iulia	1,288,451,333	30,149,682	15	7,668 K	C:\Windows\System32\svchost.exe
chrome.exe	440	Running	iulia	1,048,306,815	175,174,191	15	352,872 K	C:\Program Files\Google\Chrome\Application\chrome.exe
Discord.exe	8608	Running	iulia	322,808,694	158,670,698	47	108,736 K	C:\Users\iulia\AppData\Local\Discord\app-1.0.9003\Discor...
SearchIndexer.exe	6588	Running	SYSTEM	298,102,774	61,535,778	58	49,992 K	C:\Windows\System32\SearchIndexer.exe
chrome.exe	9264	Running	iulia	281,321,236	538,886,736	23	266,032 K	C:\Program Files\Google\Chrome\Application\chrome.exe
chrome.exe	5768	Running	iulia	246,447,657	135,990,556	16	90,756 K	C:\Program Files\Google\Chrome\Application\chrome.exe
chrome.exe	11340	Running	iulia	241,550,366	58,826,748	17	18,024 K	C:\Program Files\Google\Chrome\Application\chrome.exe
Teams.exe	13024	Running	iulia	214,663,587	42,899,308	41	137,352 K	C:\Users\iulia\AppData\Local\Microsoft\Teams\current\Tea...

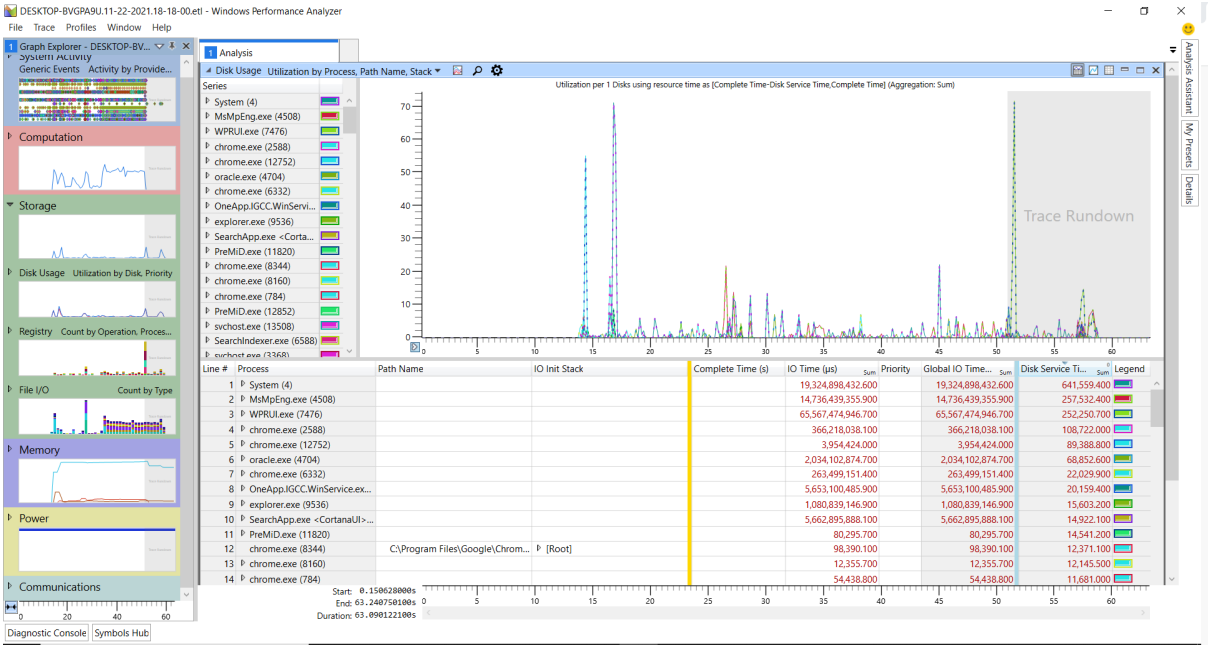
Better view:

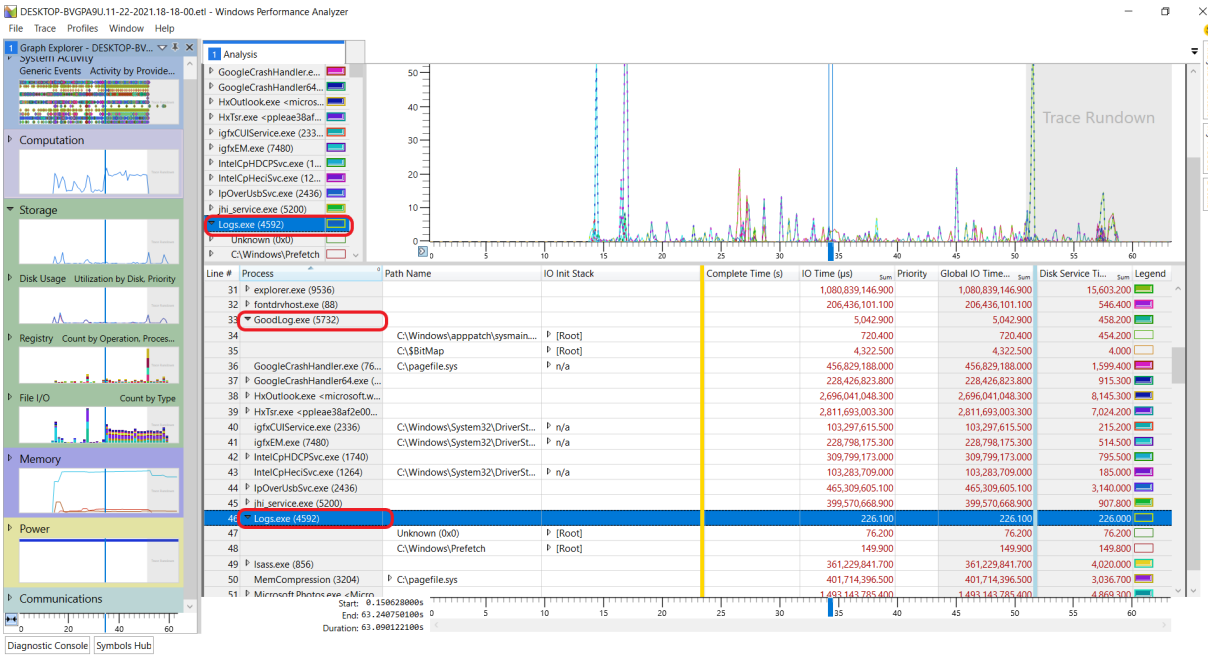
Name	PID	Status	User name	I/O read bytes	I/O write bytes
MsMpEng.exe	4508	Running	SYSTEM	36,136,406,073	6,656,465,104

MsMpEng.exe is the core process of the Windows Defender Antimalware Application. The reason why it is constantly reading/writing to my hard disk is that the executable process actively runs in the background, scans files, connections and other programs to provide protection.

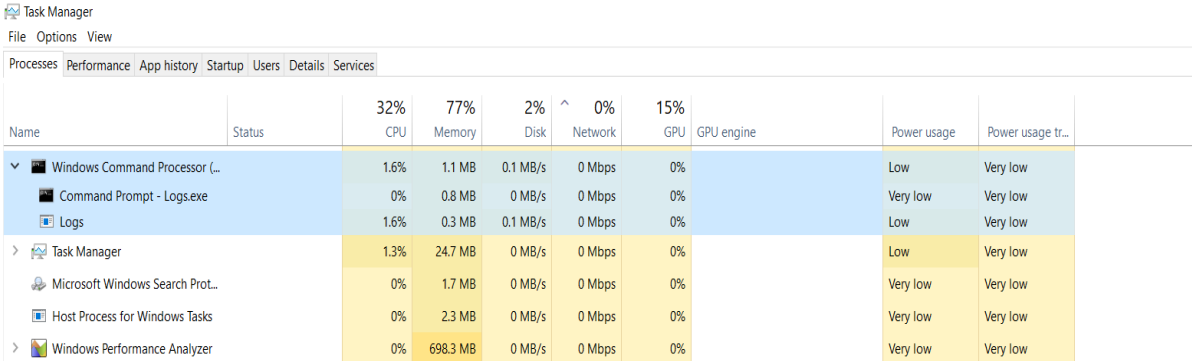
Task 02. Windows Performance Recorder & Analyzer

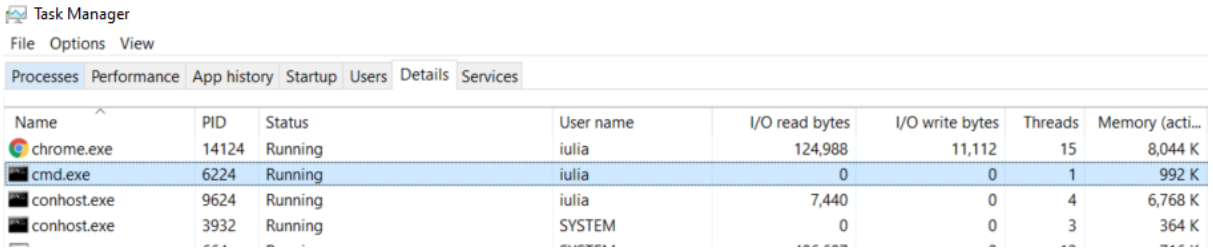
WPA Analysis for Storage:





Task Manager:

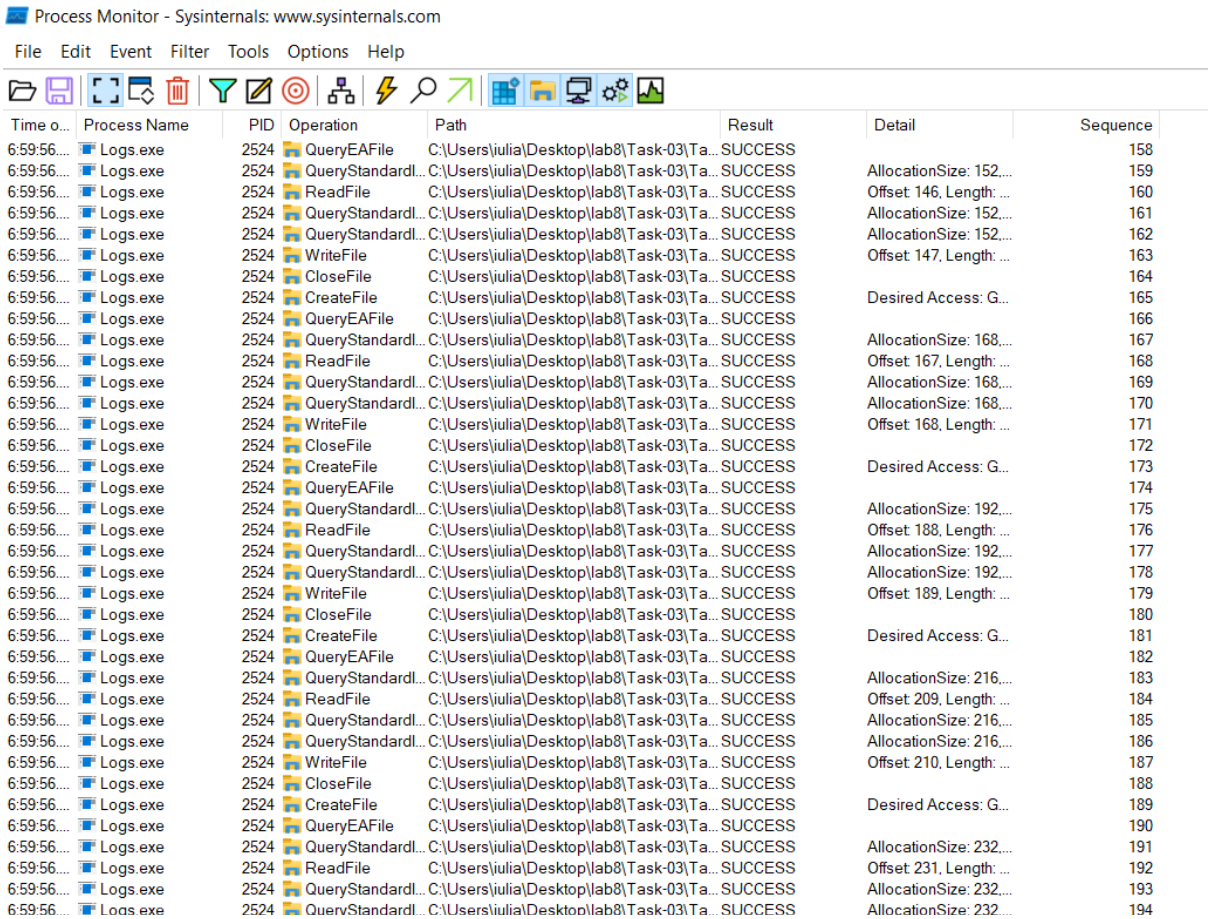




Task 03. Process Monitor

A. Checking Logging File

Logs.exe:



Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time o...	Process Name	PID	Operation	Path	Result	Detail	Sequence
6:59:56....	Logs.exe	2524	QueryEAFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS		158
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 152,...	159
6:59:56....	Logs.exe	2524	ReadFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Offset: 146, Length: ...	160
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 152,...	161
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 152,...	162
6:59:56....	Logs.exe	2524	WriteFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Offset: 147, Length: ...	163
6:59:56....	Logs.exe	2524	CloseFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS		164
6:59:56....	Logs.exe	2524	CreateFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Desired Access: G...	165
6:59:56....	Logs.exe	2524	QueryEAFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS		166
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 168,...	167
6:59:56....	Logs.exe	2524	ReadFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Offset: 167, Length: ...	168
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 168,...	169
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 168,...	170
6:59:56....	Logs.exe	2524	WriteFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Offset: 168, Length: ...	171
6:59:56....	Logs.exe	2524	CloseFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS		172
6:59:56....	Logs.exe	2524	CreateFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Desired Access: G...	173
6:59:56....	Logs.exe	2524	QueryEAFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS		174
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 192,...	175
6:59:56....	Logs.exe	2524	ReadFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Offset: 188, Length: ...	176
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 192,...	177
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 192,...	178
6:59:56....	Logs.exe	2524	WriteFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Offset: 189, Length: ...	179
6:59:56....	Logs.exe	2524	CloseFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS		180
6:59:56....	Logs.exe	2524	CreateFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Desired Access: G...	181
6:59:56....	Logs.exe	2524	QueryEAFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS		182
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 216,...	183
6:59:56....	Logs.exe	2524	ReadFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Offset: 209, Length: ...	184
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 216,...	185
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 216,...	186
6:59:56....	Logs.exe	2524	WriteFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Offset: 210, Length: ...	187
6:59:56....	Logs.exe	2524	CloseFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS		188
6:59:56....	Logs.exe	2524	CreateFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Desired Access: G...	189
6:59:56....	Logs.exe	2524	QueryEAFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS		190
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 232,...	191
6:59:56....	Logs.exe	2524	ReadFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Offset: 231, Length: ...	192
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 232,...	193
6:59:56....	Logs.exe	2524	QueryStandardI...	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	AllocationSize: 232,...	194

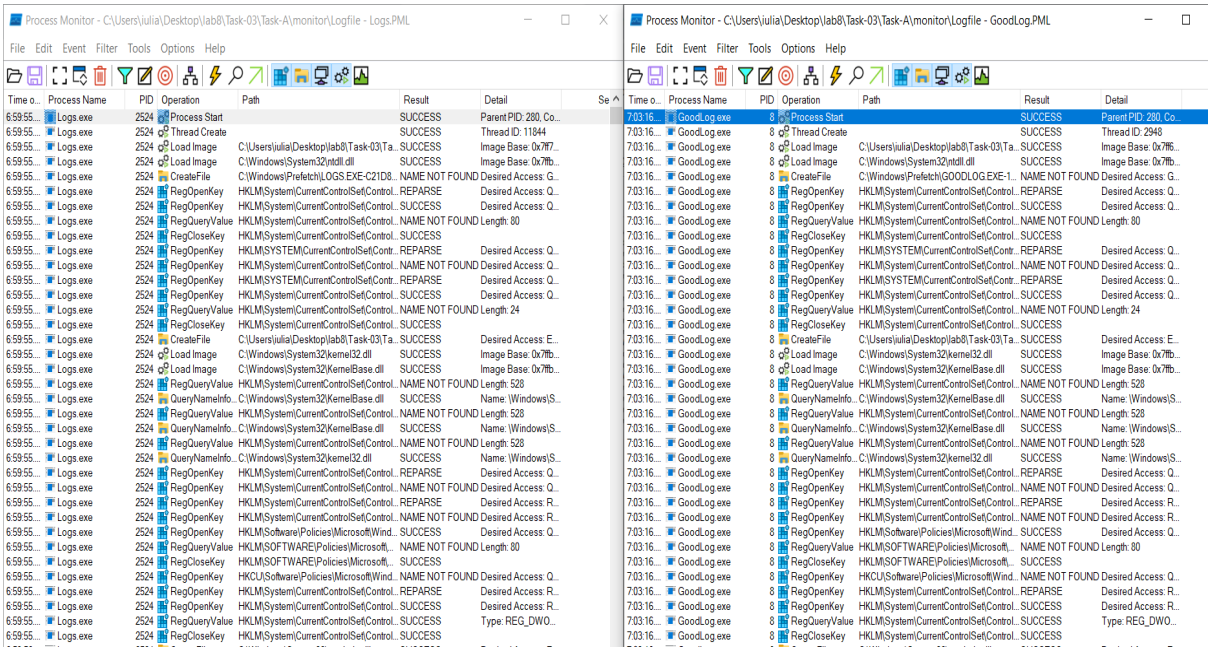
GoodLog.exe:

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time o...	Process Name	PID	Operation	Path	Result	Detail	Sequence
7:03:16....	GoodLog.exe	8	Process Start		SUCCESS	Parent PID: 280, Co...	0
7:03:16....	GoodLog.exe	8	Thread Create		SUCCESS	Thread ID: 2948	1
7:03:16....	GoodLog.exe	8	Load Image	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Image Base: 0x7ff6...	2
7:03:16....	GoodLog.exe	8	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7ffb...	3
7:03:16....	GoodLog.exe	8	CreateFile	C:\Windows\Prefetch\GOODLOG.EXE-1...	NAME NOT FOUND	Desired Access: G...	4
7:03:16....	GoodLog.exe	8	RegOpenKey	HKLM\System\CurrentControlSet\Control...	REPARSE	Desired Access: Q...	5
7:03:16....	GoodLog.exe	8	RegOpenKey	HKLM\System\CurrentControlSet\Control...	SUCCESS	Desired Access: Q...	6
7:03:16....	GoodLog.exe	8	RegQueryValue	HKLM\System\CurrentControlSet\Control...	NAME NOT FOUND	Length: 80	7
7:03:16....	GoodLog.exe	8	RegCloseKey	HKLM\System\CurrentControlSet\Control...	SUCCESS		8
7:03:16....	GoodLog.exe	8	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...	9
7:03:16....	GoodLog.exe	8	RegOpenKey	HKLM\System\CurrentControlSet\Control...	NAME NOT FOUND	Desired Access: Q...	10
7:03:16....	GoodLog.exe	8	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...	11
7:03:16....	GoodLog.exe	8	RegOpenKey	HKLM\System\CurrentControlSet\Control...	SUCCESS	Desired Access: Q...	12
7:03:16....	GoodLog.exe	8	RegQueryValue	HKLM\System\CurrentControlSet\Control...	NAME NOT FOUND	Length: 24	13
7:03:16....	GoodLog.exe	8	RegCloseKey	HKLM\System\CurrentControlSet\Control...	SUCCESS		14
7:03:16....	GoodLog.exe	8	CreateFile	C:\Users\iulia\Desktop\lab8\Task-03\Ta...	SUCCESS	Desired Access: E...	15
7:03:16....	GoodLog.exe	8	Load Image	C:\Windows\System32\kernel32.dll	SUCCESS	Image Base: 0x7ffb...	16
7:03:16....	GoodLog.exe	8	Load Image	C:\Windows\System32\KernelBase.dll	SUCCESS	Image Base: 0x7ffb...	17
7:03:16....	GoodLog.exe	8	RegQueryValue	HKLM\System\CurrentControlSet\Control...	NAME NOT FOUND	Length: 528	18
7:03:16....	GoodLog.exe	8	QueryNameInfo...	C:\Windows\System32\KernelBase.dll	SUCCESS	Name: \Windows\S...	19
7:03:16....	GoodLog.exe	8	RegQueryValue	HKLM\System\CurrentControlSet\Control...	NAME NOT FOUND	Length: 528	20
7:03:16....	GoodLog.exe	8	QueryNameInfo...	C:\Windows\System32\KernelBase.dll	SUCCESS	Name: \Windows\S...	21
7:03:16....	GoodLog.exe	8	RegQueryValue	HKLM\System\CurrentControlSet\Control...	NAME NOT FOUND	Length: 528	22
7:03:16....	GoodLog.exe	8	QueryNameInfo...	C:\Windows\System32\kernel32.dll	SUCCESS	Name: \Windows\S...	23
7:03:16....	GoodLog.exe	8	RegOpenKey	HKLM\System\CurrentControlSet\Control...	REPARSE	Desired Access: Q...	24
7:03:16....	GoodLog.exe	8	RegOpenKey	HKLM\System\CurrentControlSet\Control...	NAME NOT FOUND	Desired Access: Q...	25
7:03:16....	GoodLog.exe	8	RegOpenKey	HKLM\System\CurrentControlSet\Control...	REPARSE	Desired Access: R...	26

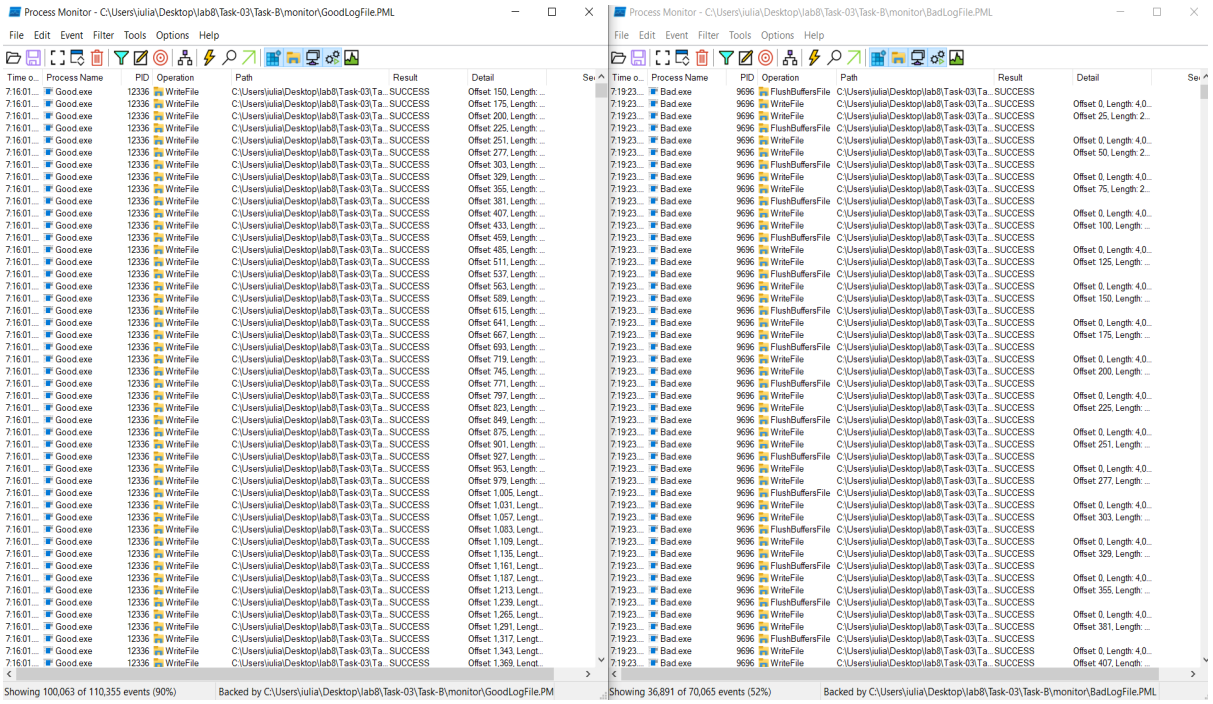
Logs.exe versus GoodLog.exe:



Logs.exe creates a file, then makes a read and a write, and after all closes the file, doing this on repeat. Meanwhile, GoodLog.exe creates it, writes multiple times, and then closes the file at the end.

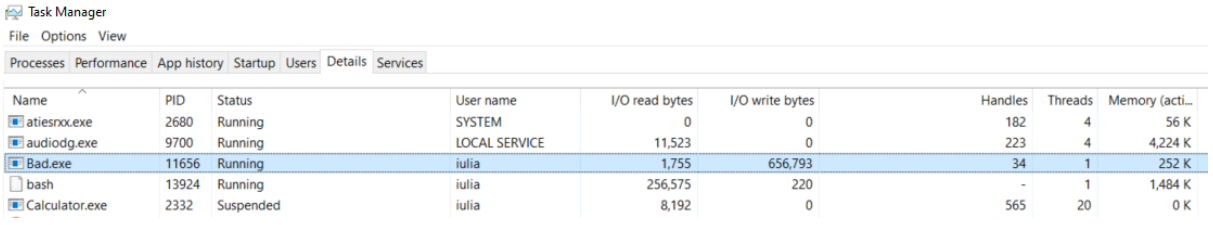
B. Investigating a Handle Leak

Good.exe versus Bad.exe



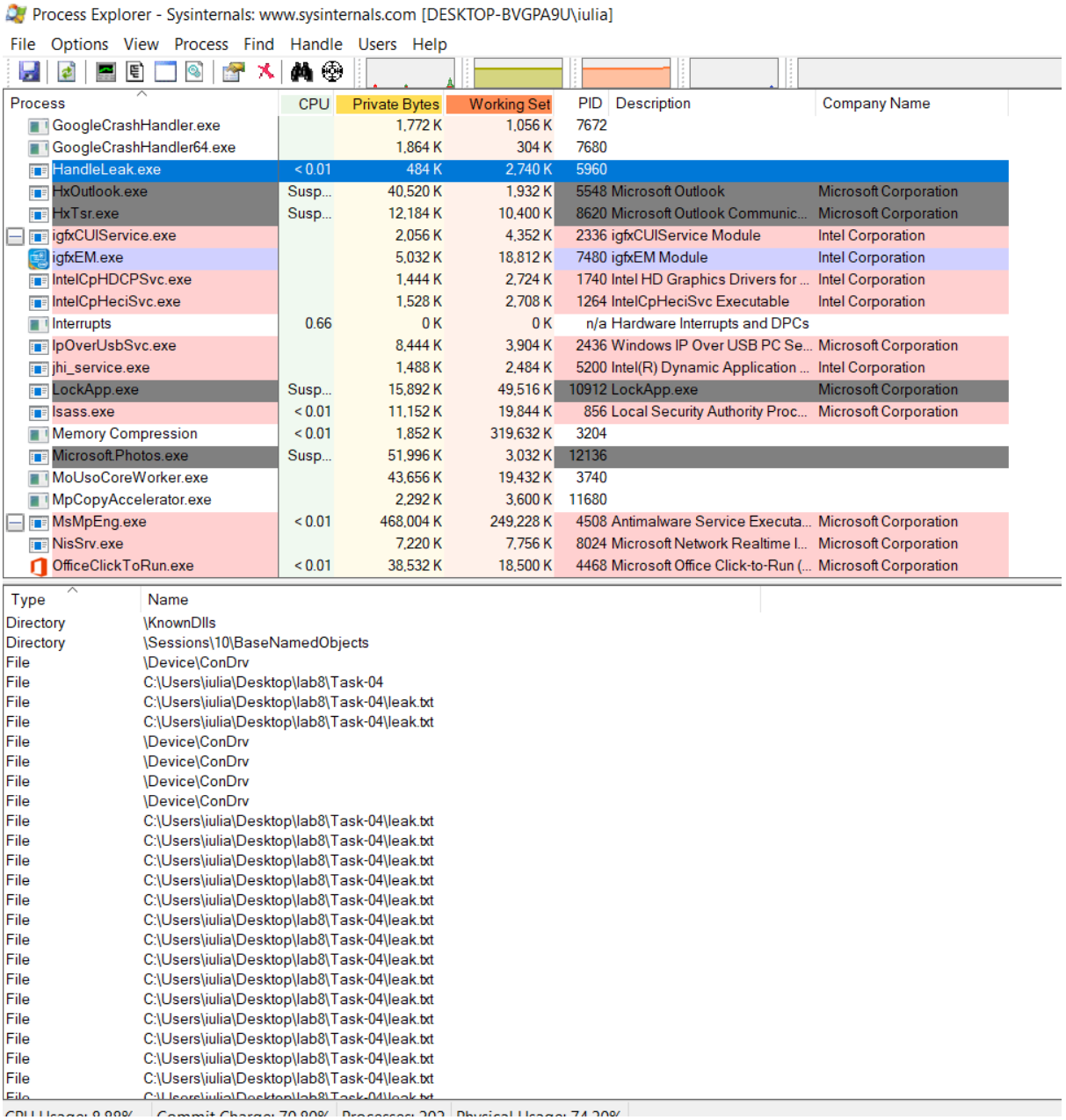
Bad.exe calls the FlushBuffersFile command every two write calls, while Good.exe writes continuously. The problem though might be the open handler.

Task Manager:

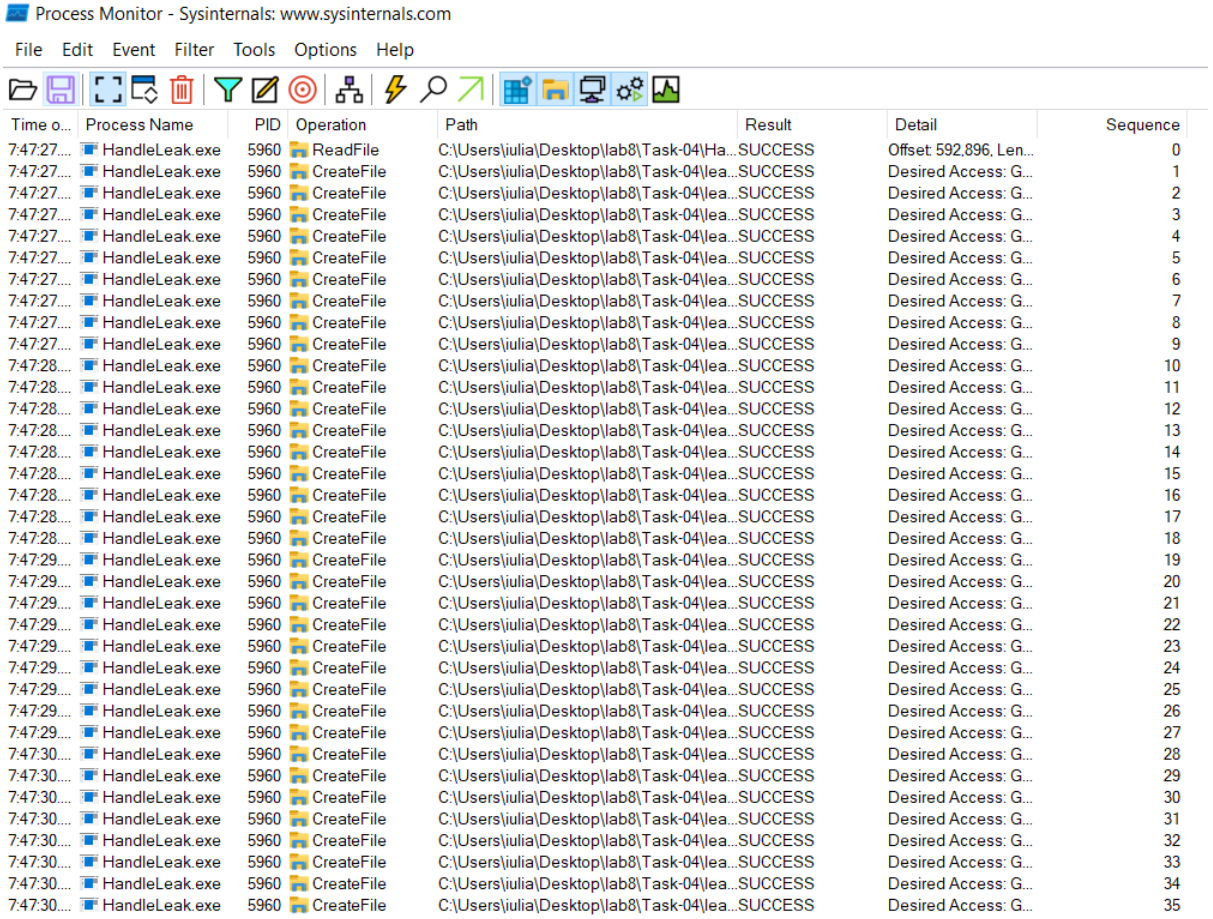


Task 04. Process Explorer

HandleLeak.exe in Process Explorer:



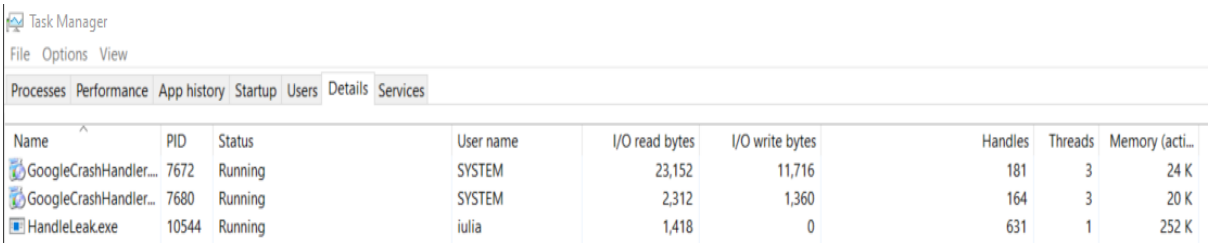
HandleLeak.exe in Process Monitor:



Time o...	Process Name	PID	Operation	Path	Result	Detail	Sequence
7:47:27....	HandleLeak.exe	5960	ReadFile	C:\Users\iulia\Desktop\lab8\Task-04\Ha...	SUCCESS	Offset: 592,896, Len...	0
7:47:27....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	1
7:47:27....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	2
7:47:27....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	3
7:47:27....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	4
7:47:27....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	5
7:47:27....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	6
7:47:27....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	7
7:47:27....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	8
7:47:27....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	9
7:47:28....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	10
7:47:28....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	11
7:47:28....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	12
7:47:28....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	13
7:47:28....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	14
7:47:28....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	15
7:47:28....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	16
7:47:28....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	17
7:47:28....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	18
7:47:29....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	19
7:47:29....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	20
7:47:29....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	21
7:47:29....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	22
7:47:29....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	23
7:47:29....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	24
7:47:29....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	25
7:47:29....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	26
7:47:29....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	27
7:47:30....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	28
7:47:30....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	29
7:47:30....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	30
7:47:30....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	31
7:47:30....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	32
7:47:30....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	33
7:47:30....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	34
7:47:30....	HandleLeak.exe	5960	CreateFile	C:\Users\iulia\Desktop\lab8\Task-04\lea...	SUCCESS	Desired Access: G...	35

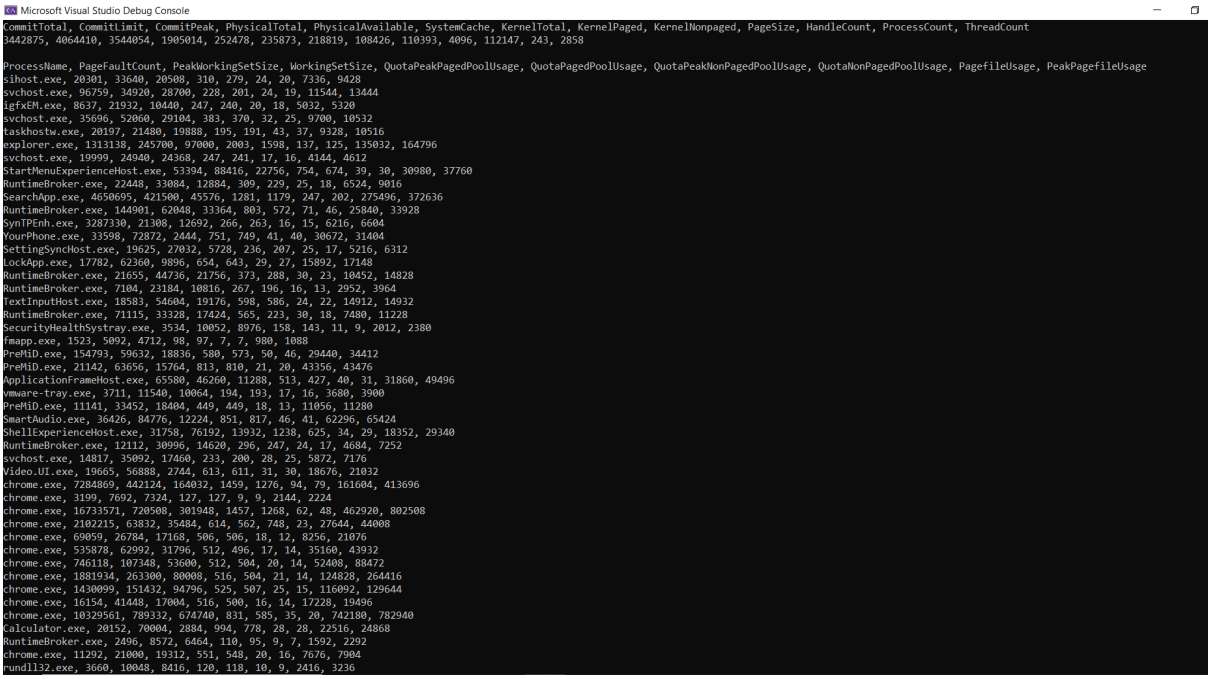
Problem is that HandleLeak.exe continuously creates handlers without closing the last one created.

HandleLeak.exe in Task Manager:



Name	PID	Status	User name	I/O read bytes	I/O write bytes	Handles	Threads	Memory (acti...
GoogleCrashHandler....	7672	Running	SYSTEM	23,152	11,716	181	3	24 K
GoogleCrashHandler...	7680	Running	SYSTEM	2,312	1,360	164	3	20 K
HandleLeak.exe	10544	Running	iulia	1,418	0	631	1	252 K

Task 05. Windows API



As the Romanian people say: “Hai ca se vede daca e!” or “Mergem pe incredere!”...
Added the code in the archive.

Tadam! 🎉🎉🎉🎉🎉🎉🎉🎉

